**Definition 2.34.** Let $A$ be a definable subset of $\mathrm{VF}^n$. The RV-*boundary* of $A$, denoted by $\partial_{\mathrm{RV}}A$, is the definable subset of $\mathrm{rv}(A)$ such that $t \in \partial_{\mathrm{RV}}A$ if and only if $t^\sharp \cap A$ is a proper nonempty subset of $t^\sharp$. The definable set $\mathrm{rv}(A) \smallsetminus \partial_{\mathrm{RV}}A$, denoted by $\mathrm{int}_{\mathrm{RV}}(A)$, is called the RV-*interior* of $A$.

**Resolved referee R032 [Typo] – Definability of the RV-boundary.**
*Original source line 945.* Insert 'definable' before 'subset of $\mathrm{VF}^n$'; without that assumption the boundary and interior need not be definable.

**Author resolution.** The author approves explicitly assuming that $A$ is definable, so that its RV-boundary and RV-interior have the stated definability.

Obviously, $A \subseteq \mathrm{VF}^n$ is an RV-pullback if and only if $\partial_{\mathrm{RV}}A$ is empty.

**Lemma 2.35.** *Let $A$ be a definable subset of $\mathrm{VF}^n$. Then $\dim_{\mathrm{RV}}(\partial_{\mathrm{RV}}A) < n$.*

*Proof.* For $n = 0$, the boundary is empty and has dimension $-\infty$. For $n = 1$, take a finite Boolean disc decomposition of $A$. A disc containing 0 is an RV-pullback, whereas a disc avoiding 0 lies in one RV-disc. A partially met RV-fiber therefore contains a member of this finite disc family that avoids 0, so there are only finitely many such fibers.

For the inductive step, fix $i \in [n]$ and put $X_i = \mathrm{pr}_{\tilde{i}}(A)$ and $T_a = \partial_{\mathrm{RV}}A_a$ for $a \in X_i$. Each $T_a$ is finite. For each $t \in T_a$, Corollary 2.21, over the parameters $(a, t)$, gives a definable point in $t^\sharp$. By compactness there are finitely many definable partial choice functions covering all such pairs $(a, t)$; order this list and use the first available choice to obtain a definable function $c(a, t) \in t^\sharp$. Set $f_i(a) = \{c(a, t) : t \in T_a\}$ if $T_a \neq \varnothing$, and $f_i(a) = \{0\}$ otherwise. Thus every fiber is finite and contains exactly one representative of each boundary tag when that boundary is nonempty; in particular $\mathrm{rv} \upharpoonright f_i(a)$ is injective.

First split $X_i$ by the pattern of its zero coordinates, and apply Corollary 2.30 on the remaining coordinates of each piece. This gives a definable finite partition $(A_{ij})_j$ of $X_i$ such that, whenever $t^\sharp \subseteq A_{ij}$, the finite set $\mathrm{rv}(f_i(a))$ is independent of $a \in t^\sharp$. The zero-pattern split allows $t$ to have $\infty$ coordinates: after deleting them, $t^\sharp$ is an open polydisc in the remaining coordinates. Write $f_{ij} = f_i \upharpoonright A_{ij}$ and let $C_{ij}(t)$ be this common finite set for $t \in \mathrm{int}_{\mathrm{RV}}(A_{ij})$.

Regard each $C_{ij}$ as its graph in $\mathrm{RV}^n_\infty$, retaining the input tags and placing its output tag in coordinate $i$. Put

$$C = \bigcup_{i,j} C_{ij}, \qquad B = \bigcup_{i,j} \{s \in \mathrm{rv}(A) : \mathrm{pr}_{\tilde{i}}(s) \in \partial_{\mathrm{RV}}A_{ij}\}.$$

Each graph $C_{ij}$ is finite over its $n-1$ input coordinates, so $\dim_{\mathrm{RV}}(C) < n$. By induction, $\dim_{\mathrm{RV}}(\partial_{\mathrm{RV}}A_{ij}) < n-1$; with only one remaining coordinate, this gives $\dim_{\mathrm{RV}}(B) < n$. Finally, for a partially met RV-polydisc, start in $A$ and change coordinates one at a time to reach a point outside $A$. The first exit identifies a boundary tag of a unary section. Its other tags either lie in some $\partial_{\mathrm{RV}}A_{ij}$, placing the full tuple in $B$, or in $\mathrm{int}_{\mathrm{RV}}(A_{ij})$, placing it in $C$. Hence $\partial_{\mathrm{RV}}A \subseteq B \cup C$, as required. □

**Resolved referee R033 [Gap] – RV-boundary proof needs uniform branches.**
*Original source line 962.* When constructing $f_{ij}$, ensure that each fiber contains exactly one representative of each boundary RV-disc; injectivity of rv on these finite fibers is a hypothesis of Corollary 2.30. The sentence about constancy should mean that the finite set $\mathrm{rv}(f_{ij}(a))$ is independent of $a \in t^\sharp$, not that rv is constant on the union of all those finite sets. The argument also depends on the preceding corollary after its parameter and multidimensional gaps are repaired. A shorter alternative is the human referee's generic-lift argument: over a model containing the parameters, a tuple of full RV-dimension has a lift realizing one complete type, so its lift cannot meet $A$ partially; state the dimension/type fact precisely if using this route.

**Author resolution.** The author approves choosing exactly one VF representative of each boundary RV tag and stating constancy of the finite RV-image set along each source RV fiber. The proof now splits zero-coordinate faces before applying Corollary 2.30, retains all input tags in the graphs, and gives the zero-dimensional base case and the coordinate-chain coverage argument.

Let $f : \mathrm{VF}^n \longrightarrow \mathrm{VF}^m$ be a definable function.

**Lemma 2.36** ([17, Lemma 8.7])**.** *There is a definable closed set $A \subseteq \mathrm{VF}^n$ with $\dim_{\mathrm{VF}}(A) < n$ such that $f \restriction (\mathrm{VF}^n \smallsetminus A)$ is continuous.*

Per the usual terminology, this lemma says that $f$ is continuous *almost everywhere*.

**Definition 2.37.** In this definition, for a nonzero tuple $x \in \mathrm{VF}^k$ we set $\mathrm{val}(x) = \min_i \mathrm{val}(x_i)$, and we set $\mathrm{val}(0) = \infty$. This minimum convention is separate from the coordinatewise valuation image used in Terminology 2.39.

For any $a \in \mathrm{VF}^n$, we say that $f$ is *differentiable at $a$* if there is a linear map $\lambda : \mathrm{VF}^n \longrightarrow \mathrm{VF}^m$ (of VF-vector spaces) such that, for any $\epsilon \in \Gamma$, if $b \in \mathrm{VF}^n \setminus \{0\}$ and $\mathrm{val}(b)$ is sufficiently large then

$$\mathrm{val}(f(a+b) - f(a) - \lambda(b)) - \mathrm{val}(b) > \epsilon.$$

It is straightforward to check that if such a linear function $\lambda$ (represented by a matrix with entries in VF) exists then it is unique and hence is called the *derivative of $f$ at $a$*.

Write $f = (f_1, \ldots, f_m)$. For $a = (a_i, a_{\tilde{i}}) \in \mathrm{VF}^n$, if the derivative of the function $f_j \restriction (\mathrm{VF} \times a_{\tilde{i}})$ at $a_i$ exists then we call it the *$ij$th partial derivative of $f$ at $a$*. Whenever $f$ is differentiable at $a$, the $(j,i)$-entry of its derivative matrix is this $ij$th partial derivative.

**Resolved referee R034 [Clarification; revised] – Valuation of a tuple and the zero increment.** *Original source line 980.* Define $\mathrm{val}(b) = \min_i \mathrm{val}(b_i)$ for a nonzero tuple, with $\mathrm{val}(0) = \infty$, and require $b \neq 0$ in the difference quotient. Otherwise the display includes the undefined expression $\infty - \infty$ at $b = 0$. Later, where valuations of tuples are taken coordinatewise (Terminology 2.39), distinguish that use from this minimum convention. The input-first indexing of partial derivatives is legitimate, but please keep the corresponding matrix convention explicit.

**Author resolution.** The author approves the minimum coordinate valuation for differentiability, with $\mathrm{val}(0) = \infty$, and excludes the zero increment from the difference quotient. This avoids $\infty - \infty$. The input-first $ij$th partial-derivative convention is retained and its derivative-matrix position $(j,i)$ is explicit. The separate boundedness convention remains coordinatewise.

**Unapplied R035 proposal for author review.** The determinant requires equal input and output VF dimensions. Generic local constancy of the output RV label supplies open fiber charts at almost every input point, rather than merely one full-dimensional fiber. On those charts the existing differentiability citation applies. The proposed proof explicitly accounts for the definable exceptional loci and the zero-dimensional case. R034 above is already approved and applied.

We differentiate functions between arbitrary definable sets by "forgetting" the RV-coordinates. More precisely, let

$$f : \mathrm{VF}^n \times \mathrm{RV}^m \longrightarrow \mathrm{VF}^{n'} \times \mathrm{RV}^{m'}$$

be a definable function. For each fixed input RV-tuple $t$, generic local constancy of definable RV-valued maps [17, Lemma 8.9] implies that, outside a definable set of VF-dimension less than $n$, the output RV-tuple is constant on a neighborhood of each input $a$. If this tuple is $s$, then $\mathrm{dom}(f_{(t,s)})$ contains a neighborhood of $a$. At such a point $(a,t)$, we define the partial derivatives of $f$ to be those of this local VF-map $f_{(t,s)}$. If $n = n'$ and all the partial derivatives exist, the *Jacobian of $f$ at $(a,t)$* is the determinant of the matrix whose $(j,i)$-entry is the $ij$th partial derivative, and is denoted by $\mathrm{Jcb}_{\mathrm{VF}} f(a,t)$.

**Lemma 2.38.** *Assume $n = n'$. The Jacobian of $f$ exists and is a continuous function almost everywhere.*

*Proof.* First remove the definable locus where the output RV-tuple is not locally constant in the input VF-coordinates. On each resulting open fiber with both input and output RV-tuples fixed, extend the VF-map by zero outside its domain and apply [17, Corollary 9.9] to its partial derivatives. Their nonexistence and discontinuity loci are definable and have VF-dimension less than $n$ in every such fiber. Retaining or projecting away RV-coordinates does not increase VF-dimension, so the combined exceptional locus still has VF-dimension less than $n$. Off this locus all partial derivatives, and hence their determinant, are continuous. In arity zero the Jacobian is the empty determinant 1. □

**Referee R035 [Gap] – Equal VF dimensions and generic RV constancy.**
*Original source line 993.* State $n = n'$ in Lemma 2.38; otherwise the determinant called the Jacobian has not been defined. Moreover, the preceding existence of one full-dimensional fiber for each input $t$ does not show that such fibers cover almost every input point. Use the standard generic local constancy of a definable map from $\mathrm{VF}^n$ to $\mathrm{RV}^{m'}$ first, and then apply the cited differentiability theorem on its open fibers. This provides the missing almost-everywhere coverage.

*Terminology* 2.39. We say that a set $I \subseteq \Gamma_\infty^n$ is $\gamma$-*bounded*, where $\gamma \in \Gamma$, if it is contained in the box $[\gamma, \infty]^n$, and is *doubly* $\gamma$-*bounded* if it is contained in the box $[-\gamma, \gamma]^n$. More generally, let $A$ be a subset of $\mathrm{VF}^n \times \mathrm{RV}_\infty^m \times \Gamma_\infty^l$ and

$$A_\Gamma = \{(\mathrm{val}(a), \mathrm{vrv}(t), \gamma) \mid (a, t, \gamma) \in A\} \subseteq \Gamma_\infty^{n+m+l};$$

then we say that $A$ is $\gamma$-*bounded* if $A_\Gamma$ is, and so on.

**Referee R036 [Clarification] – A valuation image is not a coordinate projection.**
*Original source line 1001.* The notation $A_\Gamma$ here conflicts with Notation 2.1, where it denotes projection to existing $\Gamma$-coordinates. Use a separate notation such as $\mathrm{valim}(A)$ for this coordinatewise valuation image. Also explicitly say that 'bounded' means $\gamma$-bounded for some $\gamma$, and similarly for 'doubly bounded'.

**Definition 2.40** (Contractions)**.** A function $f : A \longrightarrow B$ is rv-*contractible* if there is a (necessarily unique) function $f_\downarrow : \mathrm{rv}(A) \longrightarrow \mathrm{rv}(B)$, called the rv-*contraction* of $f$, such that

$$(\mathrm{rv} \restriction B) \circ f = f_\downarrow \circ (\mathrm{rv} \restriction A).$$

Similarly, it is res-*contractible* (respectively, val-*contractible*) if the same holds in terms of res (respectively, val or vrv, depending on the coordinates) instead of rv.

**Lemma 2.41.** *Let $U \subseteq \mathrm{RV}^k$ be a doubly bounded set and $f : U \longrightarrow \mathrm{RV}$ a definable function. Then $f(U)$ is also doubly bounded. The same holds if the codomain of $f$ is $\Gamma$.*

*Proof.* By induction on $k$, both claims are immediately reduced to showing that if $k = 1$ and $g : U \longrightarrow \Gamma$ is a definable function then $g(U)$ is doubly bounded. Then, by $C$-minimality, we may assume that $g$ vrv-contracts to a function $g_\downarrow : \mathrm{vrv}(U) \longrightarrow \Gamma$. Since definable functions in $\Gamma$ are piecewise $\mathbb{Q}$-linear, the range of $g_\downarrow$ must be doubly bounded. □

**Referee R037 [Gap] – Uniformity in the bounded RV-image lemma.**
*Original source line 1017.* The induction on $k$ must control the bounds uniformly in the remaining coordinates; boundedness on each fiber alone does not give a bound on their union. A precise route is to consider the relation of possible values $\mathrm{vrv}(f(u))$ over $\mathrm{vrv}(u) \in \Gamma^k$. Residue-field/value-group orthogonality makes each such fiber finite, uniformly; stable embeddedness and elimination in divisible